Seat Number: .....

ÖZYEĞİN UNIVERSITY
Math 104 – Summer 2019

Midterm 1

SOLUTIONS

Name: .....

Number: .....

- Please read all directions and sign the honor code below.
- The exam will be 110 minutes long and consists of 5 questions in 6 pages. Make sure you have all the pages.
- Be sure to write your exam seat number, full name, and student number above.
- Calculators and other electronic devices such as smart phones are not allowed.
- Leaving the exam room temporarily during the exam is not allowed.
- You cannot leave the exam during the first 35 minutes.
- Please raise your hand quietly if you have any questions for the instructors or for the proctors.
- Answers given without supporting work will receive no credit.

Question	Max. Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Signature:.....

Q1) [20 pts] **(Infinite sequences and series)**

(a) Determine whether the sequence $\{a_n\}$ with $a_n = \left(1 - \frac{1}{n^2}\right)^n$ converges. If so, find its limit.

Let $f(x) = \left(1 - \frac{1}{x^2}\right)^x$. Then, $\ln f(x) = \dfrac{\ln\left(1 - \frac{1}{x^2}\right)}{x}$

$$\lim_{x\to\infty} \ln f(x) = \lim_{x\to\infty} \frac{\ln\left(1-\frac{1}{x^2}\right)}{x} = \lim_{x\to\infty} \frac{2/x^3}{\left(1-\frac{1}{x^2}\right)} \quad \text{by l'Hospital's Rule}$$

$$= \lim_{x\to\infty} \frac{2}{x^3 - x} = 0$$

Hence, $\lim_{x\to\infty} f(x) = e^0 = 1 \Rightarrow \lim_{n\to\infty}\left(1 - \frac{1}{n^2}\right)^n = \boxed{1}$

and the sequence $\boxed{\text{converges}}$.

(b) Determine whether the series $\sum_{n=0}^{\infty} \frac{4 + (-1)^n}{3^n}$ converges or diverges. If it converges, find its sum.

$$\sum_{n=0}^{\infty} \frac{4+(-1)^n}{3^n} = \underbrace{\sum_{n=0}^{\infty} 4\left(\frac{1}{3}\right)^n}_{\substack{\text{Geo-series with} \\ a=4,\ r=1/3}} + \underbrace{\sum_{n=0}^{\infty}\left(\frac{-1}{3}\right)^n}_{\substack{\text{Geo-series with} \\ a=1,\ r=-1/3}}$$

$$= \frac{4}{1-1/3} + \frac{1}{1-(-1/3)}$$

$$= \boxed{\frac{27}{4}}$$

Q2) [20 pts] **(Infinite sequences and series)** Determine whether the following series converges or diverges. Show your justification and state the name of the convergence test you use.

(a) $\sum_{n=1}^{\infty} \sin \frac{1}{n}$ Limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n}$:

$$\lim_{n\to\infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = \lim_{h\to 0^+} \frac{\sin h}{h} = 1 > 0$$

Since the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges,

the series $\sum_{n=1}^{\infty} \sin \frac{1}{n}$ also $\boxed{\text{diverges}}$.

(b) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$

Let $a_n = (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$. Then, $|a_n| = \left(1 - \frac{1}{n}\right)^{n^2}$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} < 1.$$

So, by the Root Test, $\sum_{n=1}^{\infty} |a_n|$ converges.

Therefore, $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$ $\boxed{\text{converges}}$ absolutely.

Q3) [20 pts] **(Taylor and Maclaurin series)** Given the power series

$$1 - \frac{1}{2}(x-7) + \frac{1}{4}(x-7)^2 + \cdots + \left(-\frac{1}{2}\right)^n (x-7)^n + \ldots$$

(a) Find the series' radius and interval of convergence.

(b) Find the function $f(x)$ such that the Taylor series generated by f at $x = 7$ is the given series.

(a) we have $\sum_{n=0}^{\infty} \left(\frac{-1}{2}\right)^n (x-7)^n$ with $a_n = \left(\frac{-1}{2}\right)^n (x-7)^n$.

This is a geometric series with $a=1$ and $r = \frac{-1}{2}(x-7)$.

Therefore, the series converges for $|r| = \frac{1}{2}|x-7| < 1$

(Ratio test is also possible.) $\Rightarrow \boxed{5 < x < 9}$

we have the radius of convergence $\boxed{R=2}$.

(b) In the interval $5 < x < 9$, the geometric series

Converges to the sum

$$\frac{a}{1-r} = \frac{1}{1-\left[\frac{-1}{2}(x-7)\right]} = \frac{2}{x-5}$$

Therefore, $\boxed{f(x) = \frac{2}{x-5}}$ generates the given Taylor series.

Q4) [20 pts] **(Calculus with parametric curves)** Consider the parametric curve

$$x = \frac{1}{t+1}, \quad y = \frac{t}{t-1}.$$

(a) Find an equation for the line tangent to the curve at the point when $t = 2$.

(b) Find the the value of $\frac{d^2y}{dx^2}$ when $t = 2$.

(a) When $t=2$, $x = \frac{1}{2+1} = \frac{1}{3}$, $y = \frac{2}{2-1} = 2$.

So, the tangent passes through $(\frac{1}{3}, 2)$.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-1/(t-1)^2}{-1/(t+1)^2} = \frac{(t+1)^2}{(t-1)^2}$$

$$\left.\frac{dy}{dx}\right|_{t=2} = \frac{(2+1)^2}{(2-1)^2} = 9$$ is the slope of the tangent.

Therefore, $y - 2 = 9(x - \frac{1}{3}) \Rightarrow \boxed{y = 9x - 1}$

(b) $$\frac{dy'}{dt} = \frac{d}{dt}\left[\frac{(t+1)^2}{(t-1)^2}\right] = \frac{2(t+1)(t-1)^2 - 2(t+1)^2(t-1)}{(t-1)^4}$$

$$= \frac{-4(t+1)}{(t-1)^3}$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{\frac{-4(t+1)}{(t-1)^3}}{-1/(t+1)^2} = \frac{4(t+1)^3}{(t-1)^3}$$

$$\left.\frac{d^2y}{dx^2}\right|_{t=2} = \frac{4(2+1)^3}{(2-1)^3} = \boxed{108}$$

Q5) [20 pts] **(Polar coordinates)**

(a) Sketch the region inside the circle $r = 2\sin\theta$ and outside $r = 1$ by replacing the polar equations with equivalent Cartesian equations.

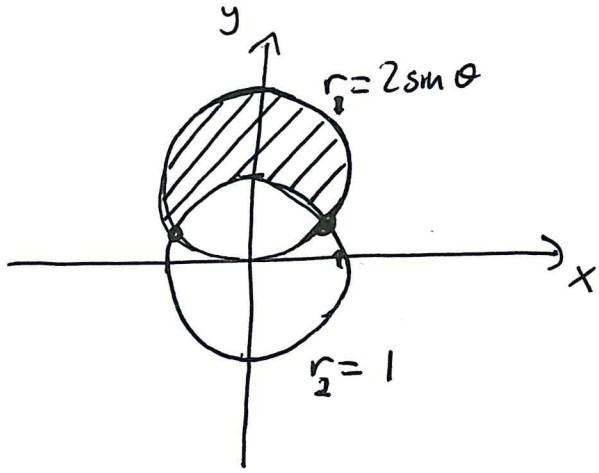
(b) Use polar integration to find the area of the region identified in part (a).

(a) $r = 1 \Leftrightarrow x^2 + y^2 = 1$

$r = 2\sin\theta \Leftrightarrow r^2 = 2r\sin\theta$

$x^2 + y^2 = 2y$

$x^2 + (y-2)^2 = 1$



(b) $r = 2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$$A = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \frac{1}{2}\left[r_1^2 - r_2^2\right] d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \frac{1}{2}\left[(2\sin\theta)^2 - (1)^2\right] d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \left(\frac{1}{2} - \cos 2\theta\right) d\theta \qquad \left(\sin^2\theta = \frac{1-\cos 2\theta}{2}\right)$$

$$= \left[\frac{1}{2}\theta - \frac{1}{2}\sin 2\theta\right]_{\pi/6}^{5\pi/6}$$

$$= \left[\frac{5\pi}{12} - \frac{1}{2}\left(-\frac{\sqrt{3}}{2}\right)\right] - \left[\frac{\pi}{12} - \frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)\right]$$

$$= \boxed{\frac{\pi}{3} + \frac{\sqrt{3}}{2}}$$